

Supply Chain Management

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Ch 13. Linking Product Availability to Profits



Chapter Overview

1. Factors Affecting the Desired Level of Product Availability
2. Evaluating the Optimal Level of Product Availability
3. Basic Managerial Levers to Improve Supply Chain Profitability
4. The Value of Speed in a Seasonal Supply Chain
5. The Value of Postponement in a Seasonal Supply Chain



13.1. Factors Affecting the Desired Level of Product Availability

- Product availability measured by cycle service level (*CSL*) or product fill rate (*fr*)
- Product availability affects supply chain responsiveness, and eventually the revenue for the supply chain
- **Trade-off** for high levels of product availability
 - Increased responsiveness and higher revenues
 - Increased inventory levels and higher supply chain costs
- **Two key factors** that influence the optimal level of product availability
 - Cost of overstocking the product, C_o
 - Cost of understocking the product, C_u



13.1. Factors Affecting the Desired Level of Product Availability

- Product availability: Optimal level – Single order
 - Example: L. L. Bean (Large online-order company selling apparel)
 - Women's red ski parka sales (for the season of November - February)
 - Procurement in advance before season
 - All customer demand and the orders to manufacturer in hundreds of parkas



13.1. Factors Affecting the Desired Level of Product Availability

- Demand distribution for parkas at L.L. Bean

i	Demand (D_i)	Probability that demand equals to D_i (p_i)	Cumulative Probability of Demand Being D_i or Less (P_i)	Probability of Demand Being Greater than D_i ($1 - P_i$)
4	400	0.01	0.01	0.99
5	500	0.02	0.03	0.97
6	600	0.04	0.07	0.93
7	700	0.08	0.15	0.85
8	800	0.09	0.24	0.76
9	900	0.11	0.35	0.65
10	1,000	0.16	0.51	0.49
11	1,100	0.20	0.71	0.29
12	1,200	0.11	0.82	0.18
13	1,300	0.10	0.92	0.08
14	1,400	0.04	0.96	0.04
15	1,500	0.02	0.98	0.02
16	1,600	0.01	0.99	0.01
17	1,700	0.01	1.00	0.00

Expected demand during a season: $\sum D_i p_i = 1,026$ units



13.1. Factors Affecting the Desired Level of Product Availability

- Product availability: Optimal level – Single order
 - Example: L. L. Bean (Large online-order company selling apparel)
 - Input Data: Per unit, purchasing cost (c) = \$45, sales price (p) = \$100 discounted price to outlet stores = \$50 (for unsold products)
 - Holding and transportation cost to outlet store = \$10, so, salvage value (s) = \$40
 - Analysis:

Expected profit from ordering 1,000 ($i = 10$) parkas:

$$\sum_{i=4}^{10} [D_i(p - c) - (1,000 - D_i)(c - s)] p_i + \sum_{i=11}^{17} [1,000(p - c)] p_i = \$49,900$$

Marginal expected profit for additional 100 units order:

$$(\$55 \times 100) \times \text{Prob}(\text{Demand} \geq 1,100) - (\$5 \times 100) \times \text{Prob}(\text{Demand} < 1,100)$$

$$\$5,500 \times 0.49 - \$500 \times 0.51 = \$2,440$$

$$\text{Thus, total expected profit: } \$49,900 + \$2,440 = \$52,340$$



13.1. Factors Affecting the Desired Level of Product Availability

- Product availability: Optimal level – Single order
 - Example: L. L. Bean (Large online-order company selling apparel)

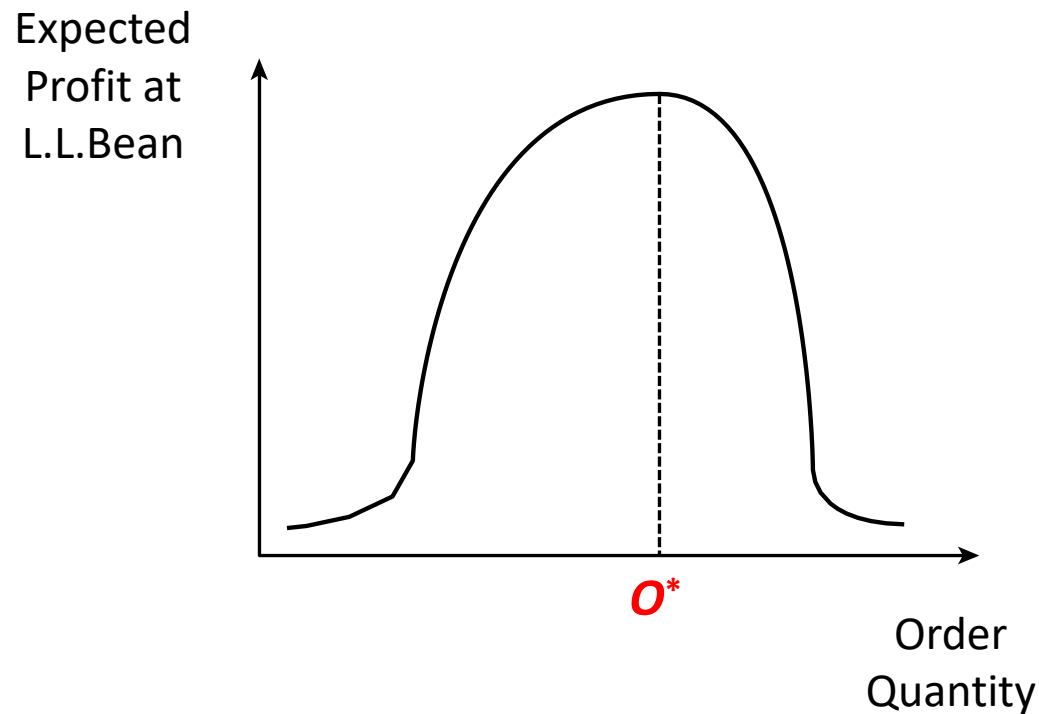
Additional Hundreds	Expected Marginal Benefit	Expected Marginal Cost	Expected Marginal Profit
1st ($O=1,100$)	$5,500 \times 0.49 = 2,695$	$500 \times 0.51 = 255$	$2,695 - 255 = 2,440$
2nd	$5,500 \times 0.29 = 1,595$	$500 \times 0.71 = 355$	$1,595 - 355 = 1,240$
3rd	$5,500 \times 0.18 = 990$	$500 \times 0.82 = 410$	$990 - 410 = 580$
4th	$5,500 \times 0.08 = 440$	$500 \times 0.92 = 460$	$440 - 460 = -20$
5th	$5,500 \times 0.04 = 220$	$500 \times 0.96 = 480$	$220 - 480 = -260$
6th	$5,500 \times 0.02 = 110$	$500 \times 0.98 = 490$	$110 - 490 = -380$
7th	$5,500 \times 0.01 = 55$	$500 \times 0.99 = 495$	$55 - 495 = -440$

Thus, the optimal order size (O^*) is 1,300 units, for which the expected profit is \$54,140 ($= \$49,900 + \$2,440 + \$1,240 + \580)



13.1. Factors Affecting the Desired Level of Product Availability

- Expected profit as a function of order quantity at L. L. Bean
 - Example: L. L. Bean (Large online-order company selling apparel)



13.1. Factors Affecting the Desired Level of Product Availability

- Product availability: Optimal level – Single order
 - Example: L. L. Bean (Large online-order company selling apparel)
 - *CSL* for optimal order
 - $CSL = \text{Prob}(\text{Demand} \leq O^* = 1,300) = 0.92$
 - *fr* for optimal order
 - If demand is 1,300 or less, a fill rate is 100% and if demand is over above 1,300, a fill rate is 1,300/D. Thus, the overall fill rate when 1,300 parkas are ordered:

$$\begin{aligned} fr &= 1 \times \text{Prob}(\text{Demand} \leq 1,300) + \sum_{D_i \geq 1,400} \left(\frac{1,300}{D_i} \right) p_i \\ &= 1 \times 0.92 + \left(\frac{1300}{1400} \right) \times 0.04 + \left(\frac{1300}{1500} \right) \times 0.02 + \left(\frac{1300}{1600} \right) \times 0.01 + \\ &\quad \left(\frac{1300}{1700} \right) \times 0.01 = 0.99 \end{aligned}$$

$$fr = 0.99 > CSL = 0.92$$



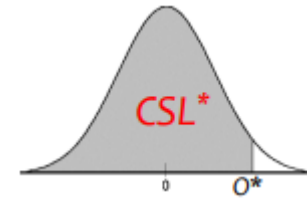
13.2. Evaluating the Optimal Level of Product Availability

- Two cases for determining the optimal *CSL*
 - Seasonal items with a single order in a season
 - Continuously stocked items
 - Demand during stockout is backlogged
 - Demand during stockout is lost
- Notations
 - p : sales retail price
 - s : salvage value
 - c : purchasing cost
 - C_o : cost of overstocking by one unit = $c - s$
 - C_u : cost of understocking by one unit = $p - c$
 - CSL^* : optimal cycle service level, i.e., *Prob(Demand during the season $\leq O^*$)*
 - O^* : corresponding optimal order size



13.2. Evaluating the Optimal Level of Product Availability

- Optimal CSL for seasonal items with a single order
 - Optimal cycle service level (based on the marginal theory)
 - Expected benefit of purchasing extra unit = $(1 - CSL^*)(p - c)$
 - Expected cost of purchasing extra unit = $CSL^*(c - s)$



$$(1 - CSL^*)(p - c) = CSL^*(c - s)$$

$$\therefore CSL^* = \frac{p-c}{p-s} = \frac{C_u}{C_u + C_o} = \frac{1}{1 + \left(\frac{C_o}{C_u}\right)} \text{ where } C_u = p - c \text{ and } C_o = c - s$$

- Optimal ordering quantity for optimal CSL

$$CSL^* = Prob(D \leq O^*) = F(O^*, \mu, \sigma) \text{ where } D \sim NORM(\mu, \sigma)$$

$$\text{So, } O^* = F^{-1}(CSL^*, \mu, \sigma) = \mathbf{NORMINV(CSL^*, \mu, \sigma)} \text{ or } \mu + z_{CSL^*} \times \sigma$$

- Optimal CSL for L. L. Bean:

$$\frac{p-c}{p-s} = \frac{55}{60} = 0.917 \approx 0.92 \quad (\text{where } p = \$100, c = \$45, s = \$40)$$



13.2. Evaluating the Optimal Level of Product Availability

- Expected profit, overstock, and understock (Order quantity O)

- Expected profit (Eq. 13.3)

$$(p - s)\mu F_s\left(\frac{O - \mu}{\sigma}\right) - (p - s)\sigma f_s\left(\frac{O - \mu}{\sigma}\right) - O(c - s)F(O, \mu, \sigma) + O(p - c)[1 - F(O, \mu, \sigma)]$$

- Expected overstock (Eq. 13.4)

$$(O - \mu)F_s\left(\frac{O - \mu}{\sigma}\right) + \sigma f_s\left(\frac{O - \mu}{\sigma}\right)$$

- Expected understock (Eq. 13.5)

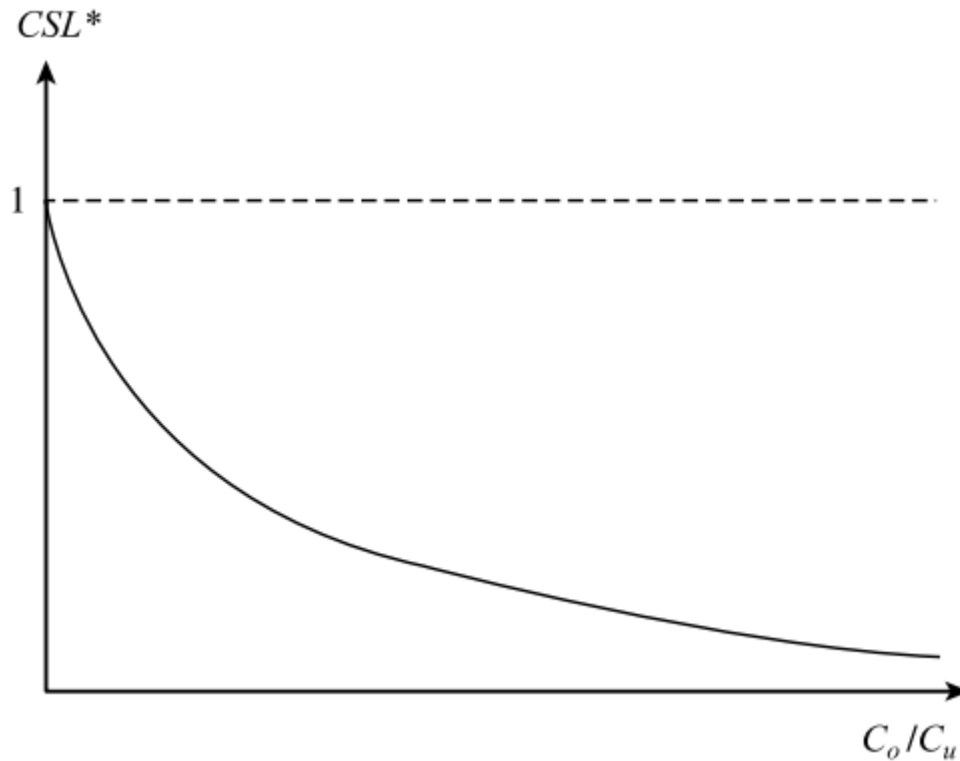
$$(\mu - O)\left[1 - F_s\left(\frac{O - \mu}{\sigma}\right)\right] + \sigma f_s\left(\frac{O - \mu}{\sigma}\right)$$

(Refer to Appendix 13C, 13D, and 13E for proof)



13.2. Evaluating the Optimal Level of Product Availability

- Impact on optimal CSL of Changing C_o/C_u



$$\therefore CSL^* = \frac{p-c}{p-s} = \frac{C_u}{C_u+C_o} = \frac{1}{1+\left(\frac{C_o}{C_u}\right)} \text{ where } C_u = p - c \text{ and } C_o = c - s$$



13.2. Evaluating the Optimal Level of Product Availability

- Example 13.1: Evaluating the optimal service level for seasonal items

- Example: Sportmart (Sporting goods store)

- Demand for ski $\sim$ NORM(350,100), purchasing cost (c) = \$100
- Retail price (p) = \$250, discounted price at the season-off = \$85
- Inventory cost (H) = \$5 (so, the actual salvage value: $s = \$85 - \$5 = \$80$)

- Understocking cost (C_u) = $p - c = \$150$; Overstocking cost (C_o) = $c - s = \$20$

$$CSL^* = Prob(D \leq O^*) = \frac{C_u}{C_u + C_o} = 0.88$$

$$Prob(D \leq O^*) = Prob\left(z \leq \frac{O^* - 350}{100}\right) = 0.88$$

$$\therefore O^* = \mu + z_{CSL^*} \times \sigma = 350 + z_{0.88} \times 100 = 468 \quad (z_{0.88} = 1.18) \text{ or}$$

$$O^* = NORMINV(CSL^*, m, s) = NORMINV(0.88, 350, 100) = 468$$

- Example 13.2: Expected profits, overstock and understock for O^* ?



13.3. Basic Managerial Levers to Improve Supply Chain Profitability

- Three basic managerial levers to increase profitability
 1. Increasing the salvage value of each unit ($s \downarrow$, so, the optimal CSL)
 - Strategy: Sale to outlet stores
 2. Decreasing the margin lost from a stockout ($p-c \downarrow$, by allowing a lower optimal CSL, and thus decreasing the excess inventory and actual shortages at the end of the season)
 - Strategies: Substitute products for shortage
 3. Decreasing the uncertainty of demand* (by decreasing both the expected overstock and the expected understock)
 - Strategy: Improved forecasts

$$\therefore CSL^* = \frac{p-c}{p-s} = \frac{C_u}{C_u + C_o} = \frac{1}{1 + \left(\frac{C_o}{C_u}\right)} \text{ where } C_u = p - c \text{ and } C_o = c - s$$



13.3. Basic Managerial Levers to Improve Supply Chain Profitability

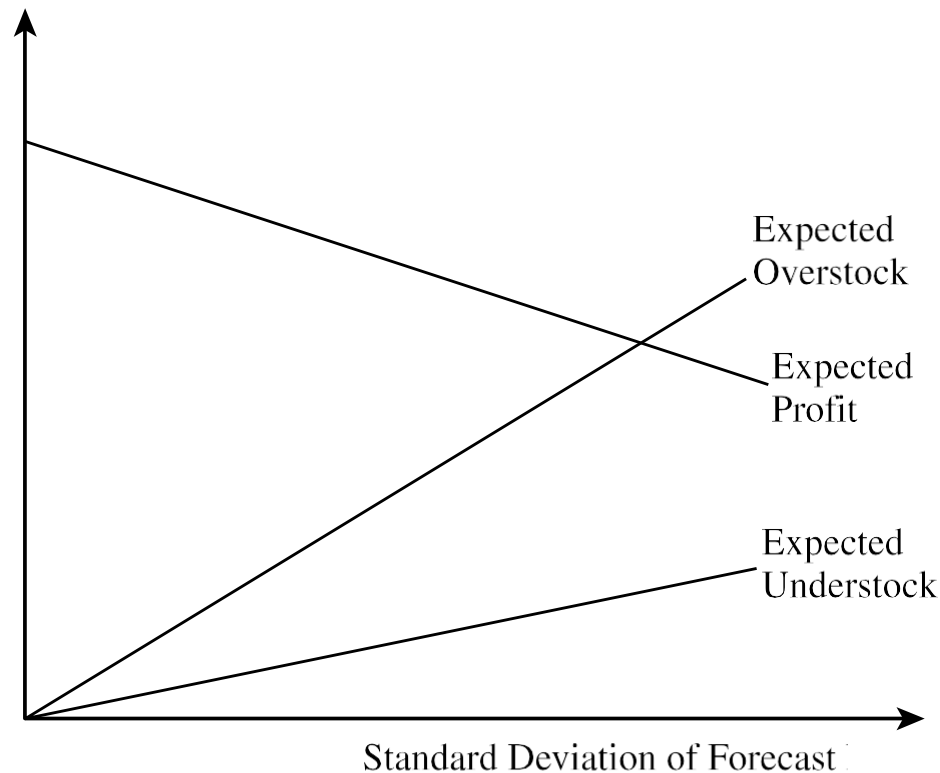
- Example 13.6: Impact of Improved Forecasts
 - Bloomingdale: Dinnerware sale only during the Christmas season
 - Demand $\sim$ NORM(350, 150) sets, $c = \$100$, $p = \$250$, $s = \$80$
 - Thus, $CSL^* = 150/(150 + 20) = 0.88$
 - Expected profits and order size at bloomingdale

Standard Deviation of Forecast σ	O^* (Eq. 13.2)	Expected Overstock (Eq. 13.4)	Expected Understock (Eq. 13.5)	Expected Profit (Eq. 13.3)
150	526	186.7	8.6	\$47,469
120	491	149.3	6.9	\$48,476
90	456	112.0	5.2	\$49,482
60	420	74.7	3.5	\$50,488
30	385	37.3	1.7	\$51,494
0	350	0	0	\$52,500



13.3. Basic Managerial Levers to Improve Supply Chain Profitability

- Variation of profit and inventories with forecast accuracy



13.4. The Value of Speed in a Seasonal Supply Chain

- **Quick response** is the set of actions a supply chain takes to reduce the replenishment lead time
- Reduced lead time results in **improved forecast accuracy**
- Typical application of quick response is **multiple orders** in one season for retail items (such as fashion clothing)
- **Benefits** of multiple orders for retailers with same product availability
 - Lower order quantity
 - Less overstock and understock quantities
 - Higher profits



13.4. The Value of Speed in a Seasonal Supply Chain

- **Multiple orders: Example**

- Two possible ordering policies (Selling season lasts 14 weeks)
 1. Supply lead time is more than 15 weeks, as a result a **single order** is placed at least 15 weeks before the beginning of the season to cover the entire season's demand
 2. Supply lead time is reduced to six weeks, as a result **two orders** are placed for the season:
 - One for *delivery at the beginning of the season* to cover the *first seven-week period*
 - The other *placed at the end of week 1* for *delivery in week 8* to cover the *second seven-week period*



13.4. The Value of Speed in a Seasonal Supply Chain

- **Two orders**

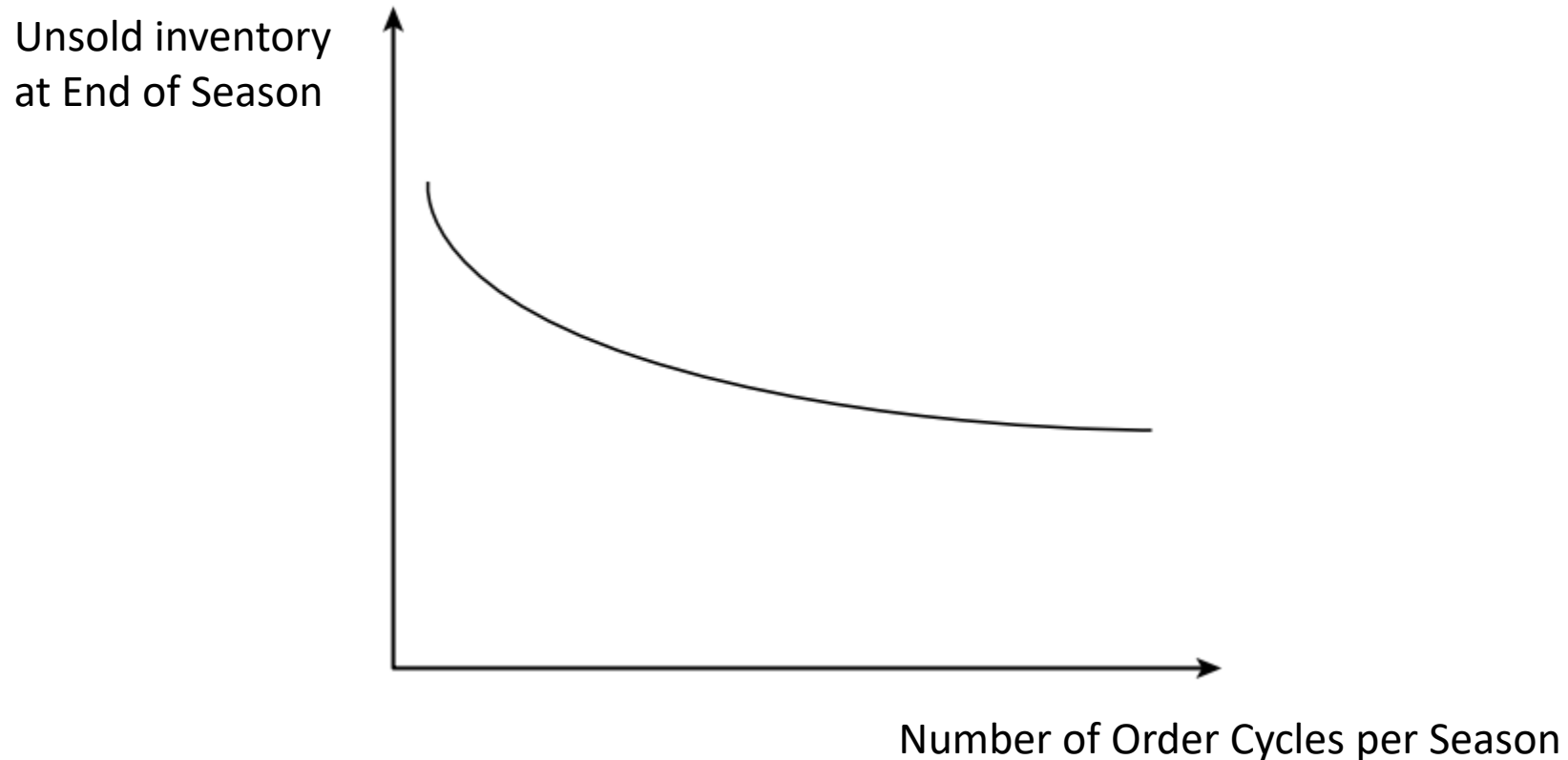
- Three important consequences

1. The **expected total quantity** ordered during the season with two orders **is less than** that with a single order for the same cycle service level (due to the accurate forecast)
2. The **average overstock** to be disposed of at the end of the sales season is **less** because a follow-up order is allowed after observing some sales
3. The **profits are higher** when a follow-up order is allowed during the sales season



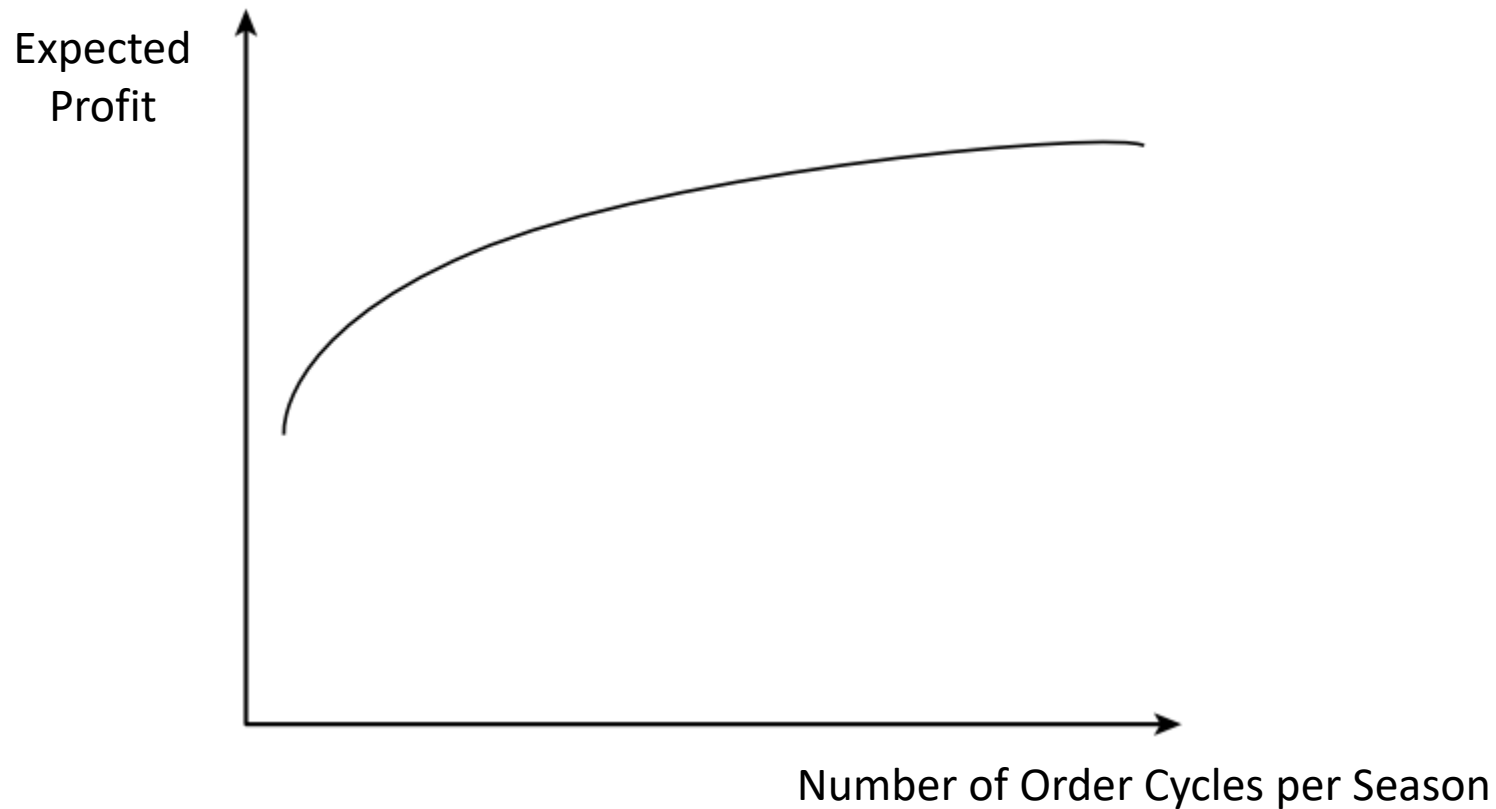
13.4. The Value of Speed in a Seasonal Supply Chain

- Leftover inventory versus number of order cycles per season



13.4. The Value of Speed in a Seasonal Supply Chain

- Leftover inventory versus number of order cycles per season



13.5. The Value of Postponement in a Seasonal Supply Chain

- **Delay of product differentiation** until closer to the time of the sale of the product, which is a powerful managerial lever to increase profitability
- Activities prior to product differentiation require **aggregate forecasts that are more accurate** than individual product forecasts
- And, individual product forecasts are required close to the time of sale with better forecast accuracy of demand
- Results in a better match of supply and demand, and so a significant increase in profits and reduction in inventories and lost sales



13.5. The Value of Postponement in a Seasonal Supply Chain

- Postponement
 - Particularly valuable if customers are willing to wait for delivery (e.g.) E-commerce (online sales)
 - Typically, the production cost associated with postponement is higher than the production cost without it (e.g.) Benetton, paint shops that mix paints
 - Value of postponement is dependent on the demand nature of products
 - **Very effective** for a firm selling a large variety of products with demand that is unpredictable, independent, and comparable in size
 - **Not very effective** for a firm selling a large variety of products with a large fraction of demand that comes from a single product

